Explaining the Temporal Structure in HERA's Redundant Calibration

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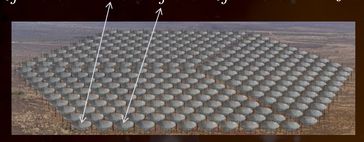
The Promise and Challenges of 21cm Cosmology

- 21 cm Cosmology uses hyperfine transition of neutral hydrogen to study the Cosmic Dawn and Epoch of Reionization.
- This can allow us to study the first stars, black holes, and galaxies via their impact on the intergalactic medium.
- **Measurements requires separation of foregrounds $10^{4\text{-}5}$ Brighter.**

21cm Requires Precision Calibration

$$V_{ij}^{obs}(\nu) = g_i(\nu) g_j^*(\nu) V_{ij}^{true}(\nu) + n_{ij}(\nu)$$



- **HERA is calibrated using redundant baselines.**
- Same baselines measure the same Visibility
- This allows solutions to antenna gains and the true Visibility of the sky.

Redundant Calibration isn't Perfect

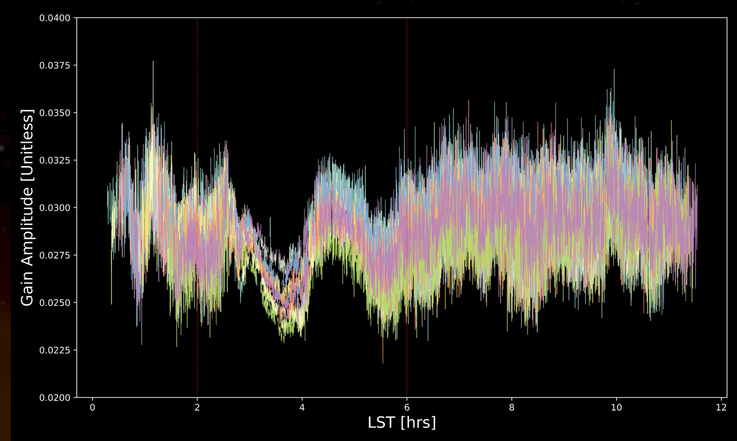
- **Non-Redundancy exists in antenna positions, beam shapes and pointing centers.**
- The system of equations has degenerate solutions.
- Bright sources can effect the gains.

Example: Spurious Temporal Structure in Measured Gains

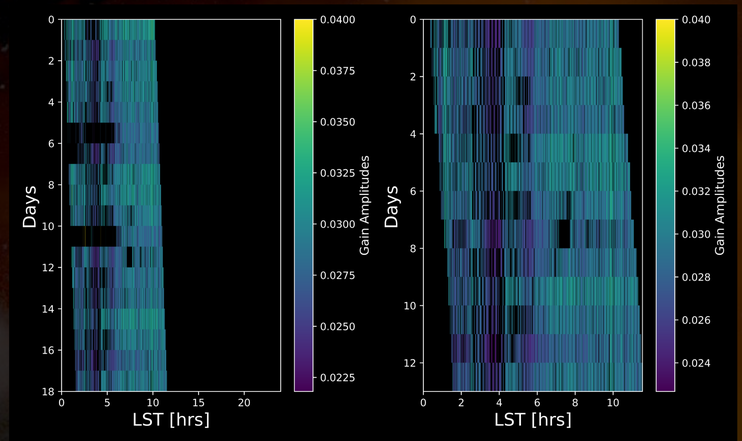
Fluctuations in the calibration solutions, repeat at the same LST everyday, implying a systematic error in calibration.

- How much of the fluctuation is calibration error, how much appears to be intrinsic?
- How do we explain the calibration error?

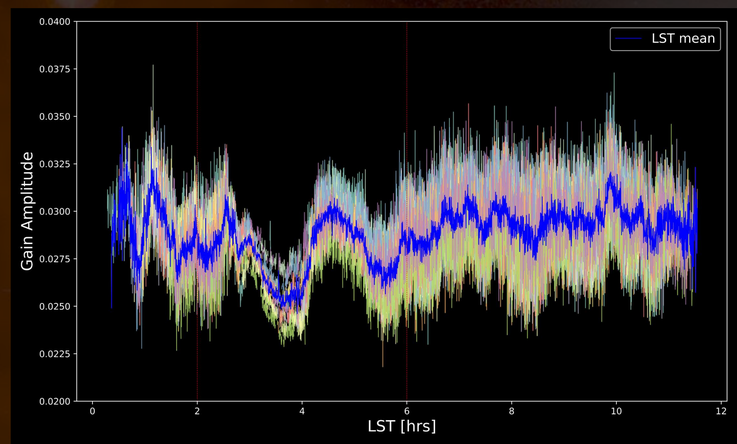
1. Gain Amplitude on Antenna 88-x



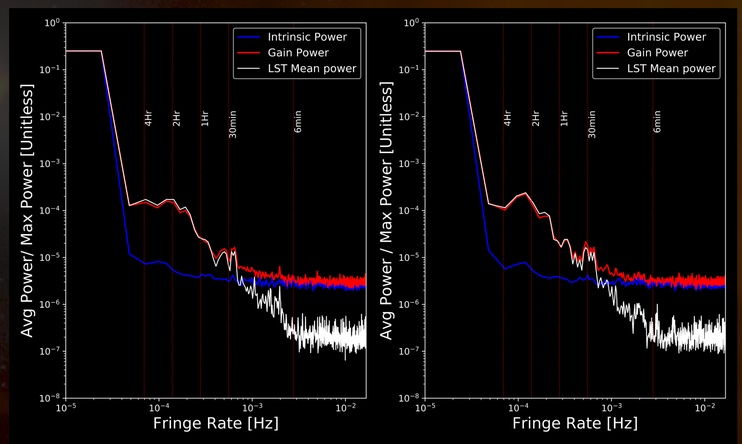
2. Extract Reliable Data



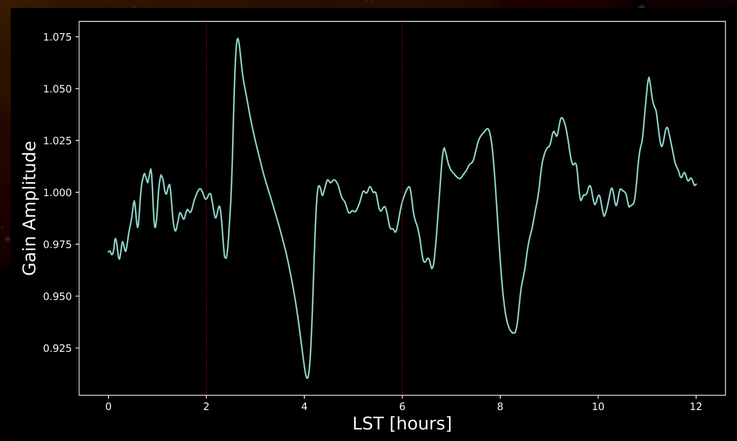
3. Use this for a Time Average



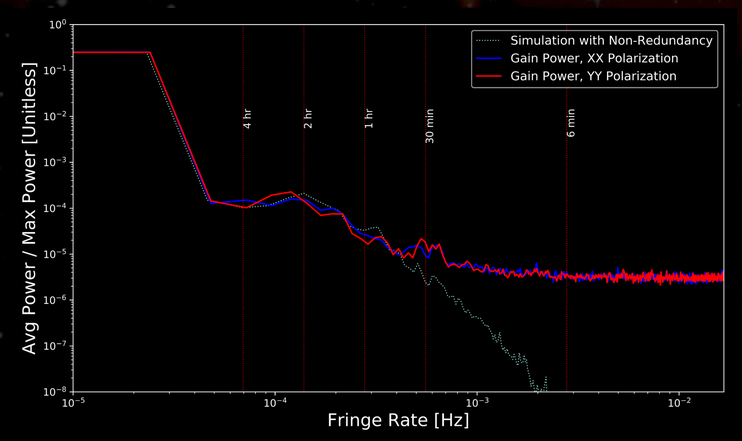
4. Compare Power Spectra, (Averaging Over Days & Antennas)



5. Simulate Non-Redundancy in Calibration

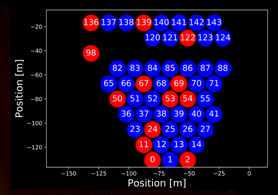


6. Compare Simulation to HERA's Data



Simulation Details

- We made a non-redundant array of antennas based off the active antennas in the data (blue active, red inactive) by randomly altering positions, beam shapes and pointing centers.



$$V_{ij}^{true}(\nu) = \int d\Omega B_{ij}(\hat{r},\nu) I(\hat{r},\nu) exp(-2\pi i \frac{\nu}{c} \mathbf{b}_{ij} \cdot \hat{r})$$

Beam shape, Baseline, True Visibility, Flux, Position

- Calculated visibility and calibrated using redundant calibration.
- Found the power spectrum of calibration gains and averaged over all active antennas.

Conclusions and Next Steps

- **Non-Redundancy in the array explains nearly all the temporal structure in the calibration solutions.**
- **Calibration should be smoothed on a longer timescale (≳4hrs) .**
- **Individual point sources (e.g. Fornax-A) in the sidelobes of the primary beam exacerbate this structure.**
- **Can they be filtered or down weighted before calibration to remove this effect? Perhaps using their structure in delay and fringe rate space?**

References

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Image Credit: Fornax-A imaged by VLA Telescope

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